



NATIONAL TEXTILE UNIVERSITY

Department of Computer Science

Lab Plan # 9: Programming Fundamentals (CSC-1075)

Basic Information			
Registration#		Name	
Total Marks		Marks Obtained	
Tools:	Dev-C++ 5.4.1		
Objectives	1. Function Overloading 2. Recursion	3. <i>Templates</i>	
Note	Solve the following problems using the concepts we have covered so far		

Functions Overloading

1. **Sum of Integers:** Create two overloaded functions named `calculateSum` to find the sum of two integers and the sum of three integers.
2. **Area Calculations:** Implement a set of overloaded functions named `calculateArea` to find the area of a rectangle, a circle, and a triangle.
3. **Print Elements:** Create overloaded functions named `printElement` to print an integer, a double, and a character.
4. **Maximum of Different Types:** Implement overloaded functions named `findMaximum` to find the maximum of two integers, two doubles, and two characters.
5. **Volume Calculations:** Create overloaded functions named `calculateVolume` to find the volume of a cube, a cylinder, and a sphere.
6. **String Operations:** Implement overloaded functions named `concatenateStrings` to concatenate two strings and three strings.
7. **Average of Different Types:** Create overloaded functions named `calculateAverage` to find the average of two integers and two doubles.
8. **Power of Numbers:** Implement overloaded functions named `calculatePower` to find the power of an integer, a double, and a long.
9. **Check Even or Odd:** Create overloaded functions named `checkEvenOdd` to check if an integer, a double, and a long are even or odd.
10. **Sorting Different Types:** Implement overloaded functions named `sortArray` to sort an array of integers, an array of doubles, and an array of characters.

Recursion

11. **Factorial Calculation:** Write a recursive function to calculate the factorial of a given positive integer.
12. **Tower of Hanoi:** Solve the Tower of Hanoi problem using recursion, where you need to move a stack of discs from one peg to another.
13. **Reverse a String:** Write a recursive function to reverse a string without using any loops.

14. Binary Search: Create a recursive function to perform a binary search on a sorted array and determine if a given element is present.

Templates

15. Create a template function that takes two arguments and returns the larger of the two. Test it with integers, floating-point numbers, and characters.
16. Write a program using a template function with two type parameters. The function should concatenate two values if they are strings, or add them if they are numeric types.
17. Create a template function to calculate the power of a number. Use double as the default type for both base and exponent.
18. Create a template function that reverses an array of any data type. Test it with integer and character arrays.